

VR Reconstruction of Amazonian Geoglyphs Using LiDAR Data

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ABSTRACT

This work presents an interactive Virtual Reality (VR) application to enhance education on Amazonian geoglyphs. Many Amazonian archaeological sites are difficult to reach and visualize as they are hidden by dense rainforest foliage. Using a combination of VR technology and LiDAR (Light Detection and Ranging) data, we generate scale models of geoglyphs without the current foliage. We use this model to create a reconstruction of the historical usage of these earthworks, based on current archaeological and anthropological interpretations. The experience is enhanced by using reconstructions of artifacts commonly found in the vicinity of these geoglyphs.

Keywords: Amazonia, Education, Geoglyphs, Virtual reality, LiDAR

1 INTRODUCTION

Over the last several decades, over a thousand geometric earthworks have been found in Southwestern Amazonia [1, 2]. These earthworks, created mostly by civilizations between 500 and 1500 years ago [2], consist of geometric shapes excavated in the ground and are typically between 100 and 350 meters wide [3]. Roads are often found radiating from the geoglyphs in multiple directions, some interconnecting geoglyphs together [1].

Substantial portions of Amazonia, where these geoglyphs are located, have not been fully explored [1-4]. While most geoglyphs have been discovered due to deforestation in the region, many geoglyphs remain covered by tropical forests [1, 2, 4]. In recent years, LiDAR scanning has been used to uncover these geoglyphs, although estimates based on LiDAR data in 2023 predict that over 10,000 geoglyph sites remain to be discovered [2].

Understanding how these geoglyphs were created and used gives crucial insight into understanding the livelihoods of Pre-Columbian civilizations in the Amazon, as well as understanding how human sociopolitical complexity in these regions may have changed over time [1-5]. However, there is no consensus on how these geoglyphs were used. Therefore, our VR application aims to further education on geoglyphs by providing simulated reconstructions of these sites when they were active. VR allows the user to be immersed in a reconstruction of a geoglyph based on LiDAR or photogrammetric data, allowing students to experience archaeological sites without the logistical challenges associated with physical visiting the geoglyphs. As John et al. [6] showed, VR applications have immense potential to immerse students and create memorable experiences.

2 METHODOLOGY

This application uses several different technologies and techniques to create reconstructions of geoglyphs. When creating the reconstructions, there are three main objectives:

- Generate a mesh from the original LiDAR data and enhance it for a VR experience.

- Create reconstructions of the historical buildings, objects, and environments around the geoglyph.
- Implement VR functionality and interactions.

2.1 Mesh Creation from LiDAR Data

2.1.1 Processing LiDAR Data

LiDAR point clouds of forested areas contain data related to trees and other vegetation. The LiDAR dataset [7] used for this work was already classified between ground and off-ground points. We removed off-ground points from the LiDAR data in R, using the LiDAR [8] package.

We use CloudCompare's CSF tool [9] to filter trees from the ground points for other data sets that have not been classified. We then import the LiDAR data containing the ground points into the CloudCompare software. Using this software, we utilize Poisson surface reconstruction to generate a mesh from the data (Fig. 1). The mesh is then used for geoglyphs reconstruction (Fig. 2).

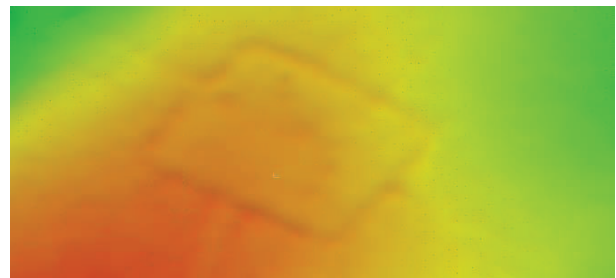


Figure 1: Mesh generated from LiDAR data.



Figure 2: Geoglyph reconstruction based on LiDAR data.

2.1.2 Mesh Editing

The original mesh generated from the Poisson reconstruction allows the geoglyph to be seen clearly from a distance. However, when viewed in VR, the large scale of the mesh makes it difficult to differentiate between average terrain and the geoglyph and its roads. Therefore, we make enhancements to the geoglyph mesh in Blender based on the dimensions of other, more pronounced geoglyphs found in the region. We scale the mesh to represent its size accurately in VR based on the dimensions of the LiDAR scan

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and remove extraneous parts of the mesh (i.e., regions in the scan that do not contain earthworks) to increase application performance.

2.2 Recreation of Historical Landscape

2.2.1 Artifacts Model Recreation

We perform 3D scans on artifacts found near the geoglyph sites using a handheld Arctec Eva scanner (see Fig.3a stone and b. basket). Based on reference images, we used modeling tool Blender to recreate objects that might be found within or near the geoglyphs. These include objects such as tools, buildings, and everyday items.

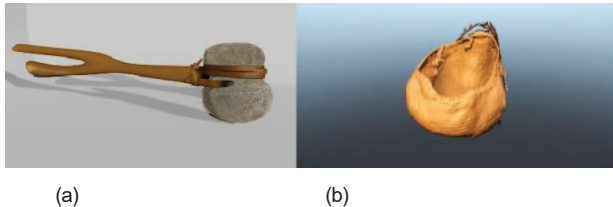


Figure 3: (a) Axe created with the 3D scanned stone found near the geoglyph and 3D modeled axe handle (b) 3D scan of a woven basket.

2.2.2 Landscape reconstruction

The meshes and models are imported into Unity game engine, where the Terrain Editor is used to place foliage and texture the mesh. Minor edits to the mesh are made with the Terrain Editor to remove jagged points and increase VR immersion. See reconstructed landscape in Fig. 4 and Fig 5.



Figure 4: Recreation of residential usage of geoglyph.



Figure 5: Road and foliage enhanced using Unity Terrain Editor

2.3 VR Interaction

Unity's XR Interaction Toolkit drives the core VR functionality and interactions of the application. The application was developed using the HTC VIVE XR Elite headset. Educational information

is placed in the scene, along with tooltips for objects the user interacts with. The user is placed in the center of the geoglyph reconstruction, with tooltips on navigating the simulation and interacting with the environment.

3 DISCUSSION & CONCLUSION

VR offers many benefits over traditional instruction. Virtual recreations of archaeological sites enable students to visualize and engage with environments that would ordinarily be extremely difficult to travel to. The combination of LiDAR data to reveal earthwork sites under dense foliage and VR technology allows for implementation of a learning experience that would ordinarily be impossible to recreate with only aerial photos of geoglyphs.

VR technology and 3D environment, however, may overwhelm the user, negatively impacting the user's attention to the educational content if not carefully designed. Furthermore, unguided VR experiences may even harm the educational experience [6]. Further steps for this work will be designed to enhance the learning experience and ensure that the VR aspect enhances the educational value with user testing. We plan to construct multiple recreations of geoglyphs to demonstrate these earthworks' wide range of hypothetical uses, which include agricultural, ceremonial, defensive, hydraulic, residential, and recreational applications. Users will be able to switch between viewing different functions of the geoglyph, enabling them to differentiate between functional explanations of the geoglyphs. Additional intended functionality for the app consists of an AR reconstruction of the geoglyphs, where students can visualize the recreated scene at the physical geoglyph sites.

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